

Ch2,3 on grade 24/30

Extra credit: 1.5

CHE 312 HW03

1. Equation of Motion

$$\rho \left(\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \right) = - \frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right) + \rho g_x$$

$$\frac{\partial v_x}{\partial t} = - \frac{\partial \psi}{\partial y} \quad \text{where } \psi = - \mu \frac{\partial^2 \psi}{\partial y^2}$$

$$\frac{\partial v_x}{\partial t} = \frac{\partial}{\partial y} \left(\frac{\partial \psi}{\partial y} \right) = \frac{\partial^2 \psi}{\partial y^2}$$

BC:

$$\psi(0, y, 0) = 0$$

$$\frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2}$$

$$\text{BC1: } v_x(0, y, 20) = v_0$$

$$\frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2}$$

$$\text{BC2: } v_x(y=20, t=20) = 0$$

using dimensionless velocity $\phi = \frac{v_x}{v_0}$, where ϕ is a function of y, t , and y

$$\frac{\partial \phi}{\partial t} = \nu \frac{\partial^2 \phi}{\partial y^2} \quad (1)$$

Since ϕ is dimensionless and a function of y, t , and y , a function η using all 3 can be constructed

$$\eta = \frac{y}{\sqrt{4\nu t}} \quad [=] \quad \frac{\sqrt{\frac{m^2}{s^2}}}{\sqrt{\frac{m^2}{s^2}}} = \frac{m}{m} = \text{dimensionless}$$

replacing variables in Eq (1) to make CDE

$$\frac{\partial \phi}{\partial t} = \frac{\partial \phi}{\partial \eta} \frac{\partial \eta}{\partial t} = \frac{1}{2} \frac{\eta}{t} \frac{\partial \phi}{\partial \eta}$$

$$\frac{\partial \phi}{\partial y} = \frac{\partial \phi}{\partial \eta} \frac{\partial \eta}{\partial y} = \frac{1}{\sqrt{4\nu t}} \frac{\partial \phi}{\partial \eta}, \quad \text{so } \frac{\partial^2 \phi}{\partial y^2} = \frac{1}{4\nu t} \frac{\partial^2 \phi}{\partial \eta^2}$$

$$\frac{1}{2} \frac{\eta}{t} \frac{\partial \phi}{\partial \eta} = \frac{1}{4\nu t} \frac{\partial^2 \phi}{\partial \eta^2}$$

$$\text{BC1: } \phi(0) = 1$$

$$\frac{\partial^2 \phi}{\partial \eta^2} + 2\eta \frac{\partial \phi}{\partial \eta} = 0 \quad (2)$$

$$\text{BC2: } \phi(20) = 0 \quad (\text{combined BC + BC2})$$

$$\text{using } \psi = \frac{\partial \phi}{\partial \eta}$$

$$\frac{\partial \psi}{\partial \eta} + 2\eta \psi = 0 \Rightarrow \frac{\partial \psi}{\partial \eta} = -2\eta \psi \Rightarrow \frac{1}{\psi} \frac{\partial \psi}{\partial \eta} = -2\eta d\eta$$

$$\ln \psi = -\eta^2 + C_1 \Rightarrow \psi = C_1 \exp(-\eta^2)$$

$$\text{re-integrating } \psi = \frac{\partial \phi}{\partial \eta}$$

$$\frac{\partial \phi}{\partial \eta} = C_1 \exp(-\eta^2) \Rightarrow \phi = \int C_1 \exp(-\eta^2) d\eta + C_2$$

$$\phi = C_1 \int_0^\eta \exp(-\eta^2) d\eta + C_2$$

08/122 absorption ...
 ... : ...

applying BCs
 $1 = C_1 \int_0^\infty \exp(-\eta^2) d\eta + C_2$
 $C_1 = \frac{1}{\int_0^\infty \exp(-\eta^2) d\eta} = \frac{\sqrt{\pi}}{2}$

$C_2 = 1$

substituting constants

$\phi(\eta) = 1 - \frac{\int_0^\eta \exp(-\eta'^2) d\eta'}{\int_0^\infty \exp(-\eta'^2) d\eta'} = 1 - \frac{\int_0^\eta \exp(-\eta'^2) d\eta'}{\frac{\sqrt{\pi}}{2}}$
 $= \frac{2}{\sqrt{\pi}} \left(\frac{\sqrt{\pi}}{2} - \int_0^\eta \exp(-\eta'^2) d\eta' \right)$

replacing $\phi = v/v_\infty$ and $\eta = \frac{y}{\sqrt{4\nu t}}$

$\frac{v}{v_\infty} = 1 - \frac{\int_0^{\frac{y}{\sqrt{4\nu t}}} \exp(-\eta'^2) d\eta'}{\int_0^\infty \exp(-\eta'^2) d\eta'} = \frac{2}{\sqrt{\pi}} \left(\frac{\sqrt{\pi}}{2} - \int_0^{\frac{y}{\sqrt{4\nu t}}} \exp(-\eta'^2) d\eta' \right)$

2. [12B.9]

From 8.2.2

$\tau_{yz} = \rho g y + C_1$ (since $\cos(0) = 1$)
 At $\tau_{yz}(0) = 0$

$0 = \rho g \delta + C_1 \Rightarrow C_1 = -\rho g \delta$

$\tau_{yz} = \rho g y - \rho g \delta$

$\mu \frac{dv_z}{dy} = \rho g (y - \delta)$

$dv_z = \frac{\rho g}{\mu} (y - \delta) dy$

$v_z = \frac{\rho g}{2\mu} \left(y^2 - 2\delta y \right) + C_2$

BC: $v_z(0) = 0 \Rightarrow C_2 = 0$

$v_z = \frac{\rho g \delta^2}{2\mu} \left(\left(\frac{y}{\delta} \right)^2 - 2 \left(\frac{y}{\delta} \right) \right)$

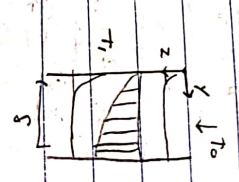
$v_{zmax} = \frac{\rho g \delta^2}{2\mu}$

Therefore

$v_z = v_{zmax} \left(2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2 \right)$

near wall, $\frac{y}{\delta}$ approaches 0. The $\left(\frac{y}{\delta} \right)^2$ term contributes very little to the velocity, so

$v_z \approx \frac{\rho g \delta}{\mu} y$



2. (continued)

$$b) \rho C_p (\mathbf{v} \cdot \nabla T) = k \nabla^2 T + \mu \nabla^2 \phi$$

$$\rho C_p \frac{\partial T}{\partial z} = k \frac{\partial^2 T}{\partial y^2}$$

$$\phi = \frac{1}{2} \leq \left[\left(\frac{\partial v_i}{\partial x_i} + \frac{\partial v_j}{\partial x_j} \right) - \frac{2}{3} (\nabla \cdot \mathbf{v}) \delta_{ij} \right]^2$$

assume $\phi_v = 0$ (low velocity and/or viscosity), Temp is linear with z

$$v_x = v_y = 0 \quad \frac{\partial T}{\partial x} = 0$$

$$\rho C_p \left(v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right) = k \left[\frac{\partial}{\partial x} \left(\frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial T}{\partial z} \right) \right]$$

$$\rho C_p v_z \frac{\partial T}{\partial z} = k \frac{\partial^2 T}{\partial y^2} = \mu \frac{\partial^2 \phi}{\partial y^2}$$

$$\text{combine with } \frac{\partial T}{\partial y} = \frac{\mu}{k} \phi$$

$$\rho C_p \mu y \frac{\partial T}{\partial z} = k \frac{\partial^2 T}{\partial y^2}$$

$$\text{use } \beta = \mu k / \rho^2 C_p g \delta$$

$$y \frac{\partial T}{\partial z} = \beta \frac{\partial^2 T}{\partial y^2}$$

c) at short time t , the heat penetration is so small compared to δ that we can assume a BC where $y \rightarrow \infty$ instead of δ .

BC1: $T(y \rightarrow 0, z) = T_0$ (fluid enters at given temp T_0 at top of fluid)

BC2: $T(y \rightarrow \infty, z) = T_0$ (small penetration, temp rise is localized to wall)

BC3: $T(z \rightarrow 0) = T_1$ (fluid at wall is at constant T_1 supplied by wall)

d) Dimensionless Analysis

$$\Theta = \frac{T - T_0}{T_1 - T_0} \quad \text{assume } \Theta \text{ is a function of } \eta(y, z, \beta)$$

$$T = \Theta(T_1 - T_0) + T_0$$

to make η dimensionless:

$$\frac{\partial T}{\partial y} = \frac{\partial \Theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} \quad \frac{\partial T}{\partial z} = \frac{\partial \Theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial z} \quad \frac{\partial^2 T}{\partial y^2} = \frac{\partial^2 \Theta}{\partial \eta^2} \cdot \left(\frac{\partial \eta}{\partial y} \right)^2$$

$$1 + \eta = \sqrt{\frac{y}{\beta z}}$$

convert η to z .

$$\frac{\partial T}{\partial z} = (T_1 - T_0) \frac{\partial \Theta}{\partial \eta} \frac{\partial \eta}{\partial z} = (T_1 - T_0) \frac{\partial^2 \Theta}{\partial \eta^2}$$

$$y \frac{\partial \Theta}{\partial z} = \beta \frac{\partial^2 \Theta}{\partial y^2} \Rightarrow y \frac{\partial \Theta}{\partial \eta} \frac{\partial \eta}{\partial z} = \beta \frac{\partial^2 \Theta}{\partial \eta^2} \frac{\partial \eta}{\partial y} \quad \frac{\partial \eta}{\partial z} = \frac{1}{\sqrt{\beta z}}$$

$$\frac{-3y \beta \frac{\partial \Theta}{\partial \eta}}{\sqrt{\beta z}} = \beta \frac{\partial^2 \Theta}{\partial \eta^2} \quad -3\eta^2 \frac{\partial \Theta}{\partial \eta} = \frac{\partial^2 \Theta}{\partial \eta^2}$$

$$\frac{\partial^3 \Theta}{\partial \eta^3} + 3\eta \frac{\partial^2 \Theta}{\partial \eta^2} = 0$$

$$\text{BC1: } \Theta(\infty) = 0 \quad (\text{comes BC1 and BC2})$$

$$\text{BC2: } \Theta(0) = 1 \quad (\text{comes BC3})$$

$$\text{Using } P = \frac{\partial \Theta}{\partial \eta}$$

$$\frac{\partial P}{\partial \eta} + 3\eta^2 P = 0 \Rightarrow \frac{\partial P}{P} = -3\eta^2 d\eta \Rightarrow \frac{1}{P} dP = -3\eta^2 d\eta$$

$$P = C_1 \exp(-\eta^3)$$

$$\text{And } P = \frac{\partial \Theta}{\partial \eta}$$

$$d\Theta = C_1 \exp(-\eta^3) d\eta$$

$$\Theta = C_1 \int_0^\eta \exp(-\eta^3) d\eta + C_2$$

Apply BCs

$$1 = C_1(0) + C_2, \quad C_2 = 1$$

$$0 = C_1 \int_0^\infty \exp(-\eta^3) d\eta + 1, \quad C_1 = \frac{-1}{\int_0^\infty \exp(-\eta^3) d\eta}$$

Therefore

$$\Theta = 1 - \frac{\int_0^\eta \exp(-\eta^3) d\eta}{\int_0^\infty \exp(-\eta^3) d\eta} = \frac{\int_0^\infty \exp(-\eta^3) d\eta - \int_0^\eta \exp(-\eta^3) d\eta}{\int_0^\infty \exp(-\eta^3) d\eta}$$

$$\Theta = \frac{\int_\eta^\infty \exp(-\eta^3) d\eta}{\int_0^\infty \exp(-\eta^3) d\eta}; \quad \text{given } \int_0^\infty \exp(-\eta^3) d\eta = \Gamma\left(\frac{4}{3}\right)$$

$$\Theta = \frac{1}{\Gamma\left(\frac{4}{3}\right)} \int_\eta^\infty \exp(-\eta^3) d\eta$$

$$\Theta = \frac{T - T_0}{T_1 - T_0} \Rightarrow T = \frac{(T_1 - T_0)}{\Gamma\left(\frac{4}{3}\right)} \int_0^\infty \exp(-\eta^3) d\eta + T_0$$

$$q = -k \frac{dT}{dy} \Rightarrow q = -k \frac{d}{dy} \left[\frac{T_1 - T_0}{\Gamma\left(\frac{4}{3}\right)} \int_0^\infty \exp(-\eta^3) d\eta + T_0 \right]$$

$$q = -k \frac{T_1 - T_0}{\Gamma\left(\frac{4}{3}\right)} \frac{d}{dy} \left[\int_0^\infty \exp(-\eta^3) d\eta \right]$$

CHE 312 HW03 (continued)

$$3. \frac{\partial T}{\partial t} = \alpha \nabla^2 T \quad \frac{\partial T}{\partial x} = 0, \quad \frac{\partial T}{\partial z} = 0$$

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial y^2}$$

Since 1 BC is in terms of heat flux at the boundary

convert PDE of T to PDE of ψ by $x = k \cdot y \cdot \frac{\partial T}{\partial y}$

$$\frac{\partial}{\partial t} \left(-k \frac{\partial T}{\partial y} \right) = \alpha \frac{\partial^2}{\partial y^2} \left(-k \frac{\partial T}{\partial y} \right)$$

$$\frac{\partial \psi}{\partial t} = \alpha \frac{\partial^2 \psi}{\partial y^2}$$

Dimensionless

$$\Theta(\eta)$$

$$\eta = \frac{y}{\sqrt{4\alpha t}}$$

$$\Theta = \frac{q_0}{q_0} \cdot \frac{1}{\sqrt{\pi}}$$

$$\xi = q_0 - \Theta q_0$$

$$BC1: \psi(0, t > 0) = q$$

$$BC2: \psi(\infty, t > 0) = 0$$

$$\frac{\partial \psi}{\partial t} = \frac{1}{2} \cdot q \cdot y \cdot (\sqrt{4\alpha t})^{-3/2} = -\frac{1}{2} \frac{q}{t}$$

$$\frac{\partial \psi}{\partial y} = \frac{1}{\sqrt{4\alpha t}}$$

$$\frac{\partial \Theta}{\partial \eta} = \frac{1}{2} \frac{q}{\sqrt{4\alpha t}} = \frac{q}{4\sqrt{\alpha t}} \frac{\partial^2 \Theta}{\partial \eta^2} \Rightarrow \frac{\partial^2 \Theta}{\partial \eta^2} + 2\eta \frac{\partial \Theta}{\partial \eta} = 0 \quad BC1: \Theta(0) = 1$$

$$\text{let } \psi = \frac{\partial \Theta}{\partial \eta}$$

$$\frac{\partial \psi}{\partial \eta} + 2\eta \psi = 0 \Rightarrow \frac{\partial \psi}{\partial \eta} = -2\eta \psi \Rightarrow \frac{1}{\psi} d\psi = -2\eta d\eta$$

$$\psi = C_1 \exp(-\eta^2), \text{ invert } \psi = \frac{\partial \Theta}{\partial \eta}$$

$$\Theta = C_1 \int_0^\eta \exp(-\tilde{\eta}^2) d\tilde{\eta} + C_2$$

apply BCs

$$1 = C_1(0) + C_2, \quad C_2 = 1$$

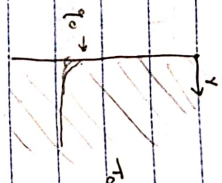
$$0 = C_1 \int_0^\infty \exp(-\tilde{\eta}^2) d\tilde{\eta} + 1, \quad C_1 = \frac{-1}{\int_0^\infty \exp(-\tilde{\eta}^2) d\tilde{\eta}} = \frac{2}{\sqrt{\pi}}$$

$$\Theta = -\int_0^\eta \exp(-\tilde{\eta}^2) d\tilde{\eta} / \int_0^\infty \exp(-\tilde{\eta}^2) d\tilde{\eta} + 1$$

$$\frac{1}{q_0} = \frac{2}{\sqrt{\pi}} \int_0^\infty \exp(-\tilde{\eta}^2) d\tilde{\eta}$$

$$\xi = \frac{2q_0}{\sqrt{\pi}} \int_0^\infty \exp(-\tilde{\eta}^2) d\tilde{\eta} = -k \frac{\partial T}{\partial y}$$

$$\frac{\partial T}{\partial y} = \frac{-q_0}{k \sqrt{\pi}} \int_0^\infty \exp(-\tilde{\eta}^2) d\tilde{\eta}$$



integrate for $T \rightarrow T_0$, with y going from $\infty \rightarrow y$ since $T(\infty) = T_0$ & $T(y) = T$

$$\int_{T_0}^T \frac{dT}{T} = \int_{\infty}^y \frac{1}{k \sqrt{\pi}} \int_0^{\infty} \eta \exp(-\eta^2) d\eta dy$$

$$\frac{1}{T} = \frac{1}{k \sqrt{\pi}} \int_{\sqrt{\pi} T}^{\infty} \exp(-\eta^2) d\eta dy + C$$

$$\frac{1}{T} = \frac{1}{k \sqrt{\pi}} \left[\int_0^{\infty} \exp(-\eta^2) d\eta \right] \int_{\sqrt{\pi} T}^{\infty} dy + C$$

$$\frac{1}{T} = \frac{1}{k \sqrt{\pi}} \left[\int_0^{\infty} \exp(-\eta^2) d\eta \right] \left[y \right]_{\sqrt{\pi} T}^{\infty} + C$$

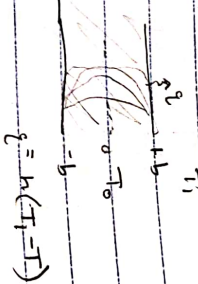
$$T = \frac{1}{k \sqrt{\pi}} \left[\int_0^{\infty} \exp(-\eta^2) d\eta \right] y + C$$

$$\frac{1}{T - T_0} = \frac{1}{k \sqrt{\pi}} \left(\sqrt{\frac{\pi}{4kt}} \exp\left(-\frac{y^2}{4kt}\right) - \sqrt{\frac{\pi}{4kt}} \int_{\sqrt{\pi} T}^{\infty} \exp(-\eta^2) d\eta \right)$$

CHE 312 HW03 (continued)

4. For given surface temperature, the analytical

solution would be given by Example 12.1-2.

$$\frac{T-T_1}{T_0-T_1} = 2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+\frac{1}{2})\pi} \exp\left[-(n+\frac{1}{2})^2 \pi^2 \kappa t / b^2\right] \cos\left((n+\frac{1}{2}) \frac{\pi y}{b}\right)$$


For given surface flux,

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial y^2}$$

$$IC: q(y, 0) = 0$$

$$BC1: q(\pm b, t > 0) = h(T_1 - T)$$

convert to PDE of q using $(-k)$ and ∂y

$$\frac{\partial}{\partial t} \left(-k \frac{\partial T}{\partial y} \right) = \kappa \frac{\partial^2}{\partial y^2} \left(-k \frac{\partial T}{\partial y} \right) \Rightarrow \frac{\partial q}{\partial t} = \kappa \frac{\partial^2 q}{\partial y^2}$$

Use separation of variables $q(y, t) = f(y) \cdot g(t)$

$$\frac{1}{f} \frac{df}{dt} = \frac{1}{g} \frac{\partial^2 f}{\partial y^2} = -c^2$$

$$\frac{df}{dt} = -c^2 g$$

$$g = A \exp(-c^2 t)$$

B must be 0 since f is symmetric with center of slab

$$c \cos y = f$$

$$\text{using } BC1: \text{at } t=0$$

$$A \exp(-c^2(0)) \cdot f \cos y = h(T_1 - T_0)$$

$$f = \frac{h(T_1 - T_0)}{A \cos y} \cdot \exp(-c^2 t) \cdot \cos y$$

$$\frac{\partial T}{\partial y} = \frac{h(T_1 - T_0)}{A \cos y} \cdot \exp(-c^2 t) \cdot \cos y$$

$$\int dT = \int \frac{h(T_1 - T_0)}{A \cos y} \cdot \exp(-c^2 t) \cdot \cos y dy$$

$$T = \frac{h(T_1 - T_0)}{A \cos y} \cdot \exp(-c^2 t) \cdot c \cdot \sin y + D$$

$$\text{using } BC2, T_1 = D$$

$$\text{using } BC1, t=0, y=b$$

$$T_0 = \frac{h(T_1 - T_0)}{A \cos y} \cdot 1 \cdot c \cdot \sin y + T_1 \Rightarrow T_1 - T_0 = \frac{h(T_1 - T_0)}{A \cos y} \cdot c \cdot \tan y$$

$$1 = \frac{h}{k} \cdot c \cdot \tan y$$